

Mohammed Zainul Abedin Kaleemi

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EDUCATION

Muffakham Jah College of Engineering & Technology (MJCET) Expected May 2028
B.E. in Mechanical Engineering, Osmania University

- **Coursework:** Scientific computing (Python, C, MATLAB), system modeling, mechanical system analysis, thermodynamics and materials science labs, embedded systems (AVR/8051), statistics, CAD (SolidWorks, AutoCAD)

EXPERIENCE

Team Robocon MJCET Nov 2024 – Present
Mechanical Head (promoted from Design Engineer and Senior Design Engineer)

- Lead the mechanical team for the ABU Robocon robot, from design and prototyping to fabrication and assembly.
- Design robot mechanisms for kinematics, strength, weight and reliability, iterating through testing.
- Produce 3D CAD models and drawings; coordinate integration with the electrical and programming teams.

Team MudBrothers (MJCET) Mar 2026 – Present
Vehicle Design Engineer, SAE BAJA

- Design parts and assemblies in CAD for the team's off-road vehicle.
- Work on subsystem design and integration, balancing weight, reliability and what can be fabricated.
- Work with the manufacturing and testing teams to revise designs; help with fabrication, assembly and testing.

Jawaharlal Nehru Technological University Hyderabad (JNTUH) Jul – Aug 2026
EV Design, Development & Manufacturing Intern

- Hybrid internship covering EV architecture, battery packs, Battery Management Systems (BMS), charging systems, MATLAB/Simulink and EV digital twins.
- Attended industry visits and expert sessions on EV development and manufacturing.

PROJECTS

S.A.F.L. (Smart Agri Four-Legged Bot) Nov 2025 – Present
Undergraduate Researcher, institution-funded R&D program

- Contributed to the mechanical design of a semi-autonomous agricultural quadruped: 3D CAD assemblies, reinforced steel chassis and articulated legs.
- Performed FEA in SolidWorks on critical components; structure validated to a minimum safety factor of 2.04.
- Designed mechanical interfaces for the actuators, soil sensing, cameras and onboard computer; assisted with Jetson Nano edge AI and sensor acquisition.
- Built and tested the prototype in field trials, including 4–6 hour field deployments.

Wearable Exoskeleton System Nov 2025 – Present
R&D project, supported under the YUKTI Innovation Challenge

- Wearable assist system to reduce strain and fatigue in manual tasks.
- Worked on the mechanical integration and assembly of a rope-driven assist using Dyneema cable, high-torque servos and 3D-printed parts.
- Controlled the servos from a headless Raspberry Pi with a Bus Servo Driver HAT.

SKILLS

CAD & Analysis: SolidWorks (Certified Professional, CSWP), AutoCAD, FEA

Fabrication & Systems: 3D printing (PLA/PETG), mechanical assembly, rapid prototyping, Linux (Arch)

Beginner level: Ansys Workbench, Git, Jetson Nano, RealSense D435, Raspberry Pi, Arduino, servo control, CNC

AWARDS

1st Place, AgriTech track, MAKEFORHYDERABAD (Titan Design Impact Movement) May 2026
All India Rank 10, IKR Go-Kart Competition (Team Asphalt MJCET)

LEADERSHIP

Club Optimus MJCET, *Technical Head* Sep 2026 – Present
IEEE IEOM MJCET, *Social Media Director* Sep 2026 – Present